

02LQD Specification and Simulation of Digital Systems
2022/02/11

1. Combinational circuit design (2 points)

A combinational circuit has a 4-bit input $M=m_0m_1m_2m_3$ and it generates a 1-bit output Z according to the following rules:

- Z is don't care if all the 4 bits of M are either at 0 or at 1
- Z is 1 if an odd number of bits of M is at 1
- Z is 0 if an even number of bits of M is at 1.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

An FSM has a 1-bit input X and a 1-bit output Z . Its goal is to remove the first 1 on every string of 1s on the input.

Example:

$X = 0\ 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 1\ 1\ 1\ 0\ 0\ 0\ 1$

$Z = 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 1\ 1\ 1\ 0\ 0\ 0\ 0$

1. Draw the state diagram of the Moore Machine (**4 points**)
2. Draw the state diagram of the Mealy Machine (**4 points**)
3. Design the **Moore** FSM resorting to the **VHDL** behavioural description style with 2 processes (**4 points**).
4. Design the **Mealy** FSM resorting in **Verilog** with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 10-bit input $Addr$, a 1-bit input Go and a 1-bit output $Finish$ and the signals (to be defined by the candidate) required to access the RAM. When the Go signal is asserted for 1 clock cycle, the functional unit reads a first memory address on its $Addr$ input. When the Go signal is asserted a second time for 1 clock cycle, the functional unit reads a second memory address on its $Addr$ input. It then sets to 0 all RAM cells whose addresses are in the range from the first address read to the second address read, boundaries included. Assume that the inputs are correct, e.g. the second address is always no less than the first one. Output $Finish$ is set to 1 when operation is over. Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
- no later than Monday February 14 5 pm**
- format of .zip archive name surname_ID (example Rossi_244123)
 - remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
 - once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.